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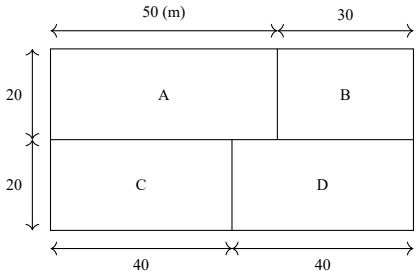
DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 25th May 2026	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Facilities Planning And Design (Professional Core Elective)		**Course Code : IM365TDA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	State the main objectives of material handling?	(02)	1	1
2.	What is meant by unit load design?	(02)	2	2
3.	Enumerate the importance of flexibility in material handling systems.	(02)	3	3
4.	What is the role of conveyors in material handling?	(02)	2	4
5.	State two objectives of computer aided layout techniques.	(02)	2	3
	Part – B			
1.	Discuss the scope of material handling in modern industries.	(10)	2	2
2.	Explain movement, time, quantity and space of a material handling system	(10)	3	2
3.	Explain the following cost components used in estimating material handling cost: i. Equipment cost ii. Storage & Inventory Costs iii. Damage/Wastage Costs	(10)	3	3
4.	Explain the safety considerations to be followed in material handling operations.	(10)	3	4

5.	Consider the following data with unit cost matrix. Use CRAFT pair-wise interchange technique to obtain the desirable layout. Total number of departments: 4 Total number of interchangeable departments: 4 Initial Layout:	(10)	2	4



Area of Departments (Scale 1000 sq units = 1 sq unit):

Department	A	B	C	D
Area (sq. units)	10	6	8	8

Flow Matrix:

From\To	A	B	C	D
A	–	2	4	4
B	1	–	1	3
C	2	1	–	2
D	4	1	0	–

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 25-05-2026	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Imagine you are going to watch an IPL cricket match at Namma Bengaluru's very own Chinnaswamy Cricket stadium, list 4 aspects that you associate regarding service experience.	2	1	1
2.	Assume you get in to an "IT enabled service" (ITES) industry in the next five years. Comment on how you intend to use Service as a competitive advantage.	2	4	3
3.	Assume you are being interviewed for your short but successful career in the field of "Service operations management". One of the questions from a young enthusiast is "How to develop a culture of service quality in an organization?" Comment any two aspects.	2	4	2
4.	Some of you have already started preparing for campus placements. In a year from now, many of your juniors would want your suggestions **(If placed)** and help in many matters of campus placements. To cater to this need and based on some of the requirement analysis that you have done, you wish to start a service and name it as "**Placement war room**" for engineering graduate students. Now answer the following questions. 1. Will it be online or offline and why? 2. What are the unmet needs of graduate students that your service will fulfil?	2	4	4
5.	If your "**Placement war room**" idea clicks and is appreciated by your initial target audience. Identify any 2-success metrics of your service? Also, comment on the possible perception expectation gap for your service.	2	4	3
	Part – B			
1	Discuss the role of technology in developing the customer Experience. Cite two examples of services where technology has been used effectively to enhance customer experience.	10	3	4
2	The nature of supply chains can be very messy and complex. Describe the key issues service operations managers should focus on with a key focus on managing through intermediaries.	10	3	4
3	Write short notes on a) Service level agreements (SLA) b) Difference between SLA and Operational Level Agreements (OLA)	5*2=10	2	3
4	Envisage/visualize an experience map for a customer visiting a typical dental clinic.	10	4	2

5	One of the key decisions for procurement function is choosing/selecting a supply partner and particularly for those who handle critical elements of supply chain. Discuss the criteria for supplier selection excluding Cost and Quality.	10	3	4

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 25/05/2026	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Design of Experiments		**Course Code : IM365TDD**

Instructions to Students:
1. Answer all questions
2. Use of statistical tables permitted

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define "Sparsity of Effects Principle."	2	1	CO1
2.	If you have a 2^5 design, how many total treatment combinations are required for a single replicate?	1	1	CO1
3.	Compare the 2^2 design and 3^2 designs in terms of modeling.	2	1	CO2
4.	What is "Confounding" in factorial experiments?	2	2	CO2
5.	Define "Aliasing" in fractional factorial designs.	1	2	CO2
6.	Evaluate the risk of using a Resolution III design for a process where two-factor interactions are known to be significant.	2	2	CO2
	Part – B			
1	An engineer is interested in the effects of cutting speed (A), tool geometry (B), and cutting angle (C) on the life (in hours) of a machine tool. Two levels of each factor are chosen, and three replicates of a 2^3 factorial design are run. The results are as follows:	20	2	CO4

A	B	C	Treatment Combination	Replicates		
				I	II	III
−	−	−	(1)	22	31	25
+	−	−	a	32	43	29
−	+	−	b	35	34	50
+	+	−	ab	55	47	46
−	−	+	c	44	45	38
+	−	+	ac	40	37	36
−	+	+	bc	60	50	54
+	+	+	abc	39	41	47

i. Estimate the factor effects. Which effects appear to be large?
ii. Use the analysis of variance to confirm your conclusions for part (i).

<table>
<tr><td>2</td><td>The effects of developer strength (A) and development time (B) on the density of photographic plate film are being studied. Three strengths and three times are used, and four replicates of a 3^2 factorial experiment are run. The data from this experiment follow. Analyze the data using the standard methods for factorial experiments.
<table>
<tr><td colspan="2" rowspan="2"></td><td colspan="12">Development Time (minutes)</td></tr>
<tr><td colspan="4">10</td><td colspan="4">14</td><td colspan="4">18</td></tr>
<tr><td rowspan="3">Developer Strength</td><td>1</td><td>0</td><td>2</td><td>1</td><td>3</td><td>2</td><td>5</td><td>4</td><td>4</td><td>2</td><td>3</td><td>2</td><td>5</td></tr>
<tr><td>2</td><td>4</td><td>6</td><td>6</td><td>8</td><td>9</td><td>10</td><td>5</td><td>4</td><td>2</td><td>4</td><td>6</td><td>7</td></tr>
<tr><td>3</td><td>7</td><td>7</td><td>5</td><td>7</td><td>7</td><td>8</td><td>5</td><td>7</td><td>10</td><td>8</td><td>9</td><td>12</td></tr>
</table></td><td>20</td><td>3</td><td>CO4</td></tr>
<tr><td>3</td><td>List and explain the three key ideas / principles that are to be implemented for the successful use of fractional factorial designs.</td><td>10</td><td>2</td><td>CO2</td></tr>
</table>

Academic year 2025-2026 (EVEN Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**22 May 2026**	**Maximum Marks**	**10 + 50**
Course Code	**IM266TEQ**	**Duration**	**120 Mins**
Sem	**VI Semester**	**CIE – II**	
ELEMENTS OF FINANCIAL MANAGEMENT (Institutional Elective)			

Note:
1. Answer all the Questions.
2. Use of Interest factor tables permitted.

PART – A

S No	Questions	M	BT	CO
1	Distinguish between the terms, Risk and Return in Financial Management.	2	2	2
2	What is meant by "Yield to Maturity" (YTM) of a bond.	2	1	2
3	Mention the three key inputs required to perform the valuation of an Asset.	2	2	2
4	What is meant by Required Rate of Return in the context of valuation of securities.	2	2	2
5	Distinguish between the terms Market Risk and Unique Risk.	2	2	2

PART – B

S No	Questions	M	BT	CO
1.a	Explain why a bond sells at a premium or discount.	5	4	2
b	A 5-year bond, coupon rate 8%, par value ₹1,000, required return 10%. Find bond price.	5	3	2
2.a	A bond has a par value ₹1,000, coupon rate 12%, maturity 5 years, current price ₹950. Compute the Yield to Maturity (YTM).	5	3	2
b	A company is expected to pay a dividend of ₹5 next year. Required rate of return is 12%. Expected price after one year is ₹55. Calculate the current value of the share.	5	3	2
3.a	A share is priced at ₹80. Dividend expected next year is ₹6. Growth rate is 4%. Find the required rate of return.	5	4	2
b	Analyse how diversification in investment portfolio reduces risk.	10	3	2
4.	An asset has returns of 10%, 15%, and 20% with probabilities 0.2, 0.5, and 0.3. Find expected return and also the variance of return.	10	3	2
5.	Enumerate the steps involved in Capital Budgeting Process.	10	2	3

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Department of Civil Engineering

Date: 26-05-2026	CIE – II	Max. Marks: 60
Semester: VI	UG	Duration: 2 Hrs
Course Title: Intelligent Transport Systems		Course Code: CV266TEE

Quiz

S No	Questions	M	BT	CO
1.	Differentiate between logical architecture and physical architecture in ITS.	2	2	2
2.	State the need for ITS architecture in urban areas.	2	1	2
3.	Mention any four communication tools used in ITS.	2	1	3
4.	Explain the role of traveller information systems in ITS.	2	2	3
5.	List the various detection and identification methods used in ITS.	2	1	3

Test

S No	Questions	M	BT	CO
1.	Describe equipment packages and market packages in ITS architecture with suitable examples.	10	3	2
2.	Explain the ITS architecture with neat sketches of logical architecture and physical architecture.	10	3	2
3.	Explain various detection, identification, and data collection methods used in Intelligent Transportation Systems.	10	3	3
4.	Explain the development of Traffic Management Systems and the functions of a Traffic Management Centre (TMC).	10	3	3
5.	Explain the concepts of Advanced Vehicle Control Systems and Advanced Public Transport Systems.	10	3	3

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 6th June 2025	**CIE – 2**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: FACILITIES PLANNING AND DESIGN		**Course Code : IM365TDA**

Note: Answer all the Questions.

PART A

Q No.1	Questions	M	BT	CO
1.1	Given station times: • Station 1: 5 min • Station 2: 4 min • Station 3: 6 min Find the bottleneck.	01	2	1
1.2	Total task time = 20 minutes, Number of workstations = 4, Cycle time = 6 minutes Find line efficiency.	02	1	1
1.3	Calculate Throughput Time. Given: • Station A: 2 min • Station B: 3 min • Station C: 4 min What is the total throughput time for one product?	02	2	1
1.4	If a production unit has limited floor space and needs to allow a single operator to handle multiple machines easily, which layout (S-flow or U-flow) is better?	01	1	1
1.5	Define a unit load as per Tanchoco et al.	02	2	1
1.6	List any two basic methods of moving the unit load according to Apple.	01	1	1

Part –B

Sl. No	Questions	M	BT	CO
1	State and explain the different types of material flow with suitable examples.	10	2	1

<table>
<tr><td>2</td><td>A factory is designing a new assembly line for producing electric scooters. The production involves 5 sequential tasks with the following task times (in minutes):
<table><tr><th>Task</th><th>Time (min)</th></tr><tr><td>A</td><td>4</td></tr><tr><td>B</td><td>6</td></tr><tr><td>C</td><td>5</td></tr><tr><td>D</td><td>3</td></tr><tr><td>E</td><td>2</td></tr></table>The management is considering either a Straight Line Flow layout or a U-Flow layout. Tasks:
a) Calculate the total time required to complete one unit.
b) Identify the bottleneck in the process.
c) If the total cycle time is 6 minutes and there are 3 workstations, compute the line efficiency.
d) Which layout (S-Flow or U-Flow) would you recommend if space is limited and one operator needs to handle multiple stations? Justify your answer</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>3</td><td>What are the Factors for Consideration in Planning Material Flow</td><td>10</td><td>2</td><td>2</td></tr>
<tr><td>4</td><td>Enumerate and elaborate the steps in material handling system design.</td><td>10</td><td>1</td><td>2</td></tr>
<tr><td>5</td><td>Explain the guidelines for estimating the material handling costs in practice.</td><td>10</td><td>1</td><td>1</td></tr>
</table>

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 6-6-2025	**CIE – 2**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Think of a time when you recently received poor service. If service operations are so important, identify reasons why poor service is delivered sometimes?	2	2	1
2.	Assume you get in to an "IT enabled service" (ITES) industry in the next five years. Comment on how you intend to use Service as a competitive advantage.	2	4	3
3.	What is the service concept of "Cyber nurseries".	2	2	2
4.	Comment on the importance of service process design.	2	3	4
5.	From your experience, give two examples where technology is used to enhance customer experience.	2	2	2
	Part – B			
1	Discuss the role of technology in developing the customer Experience. Cite two examples of services where technology has been used effectively to enhance customer experience.	10	3	4
2	The nature of supply chains can be very messy and complex. Describe the key issues service operations managers should focus on with a key focus on managing through intermediaries.	10	3	4
3	Write short notes on a) Service level agreements (SLA) b) Difference between SLA and Operational Level Agreements (OLA)	10	2	3
4	Visualize an experience map for a customer visiting a typical dental clinic.	10	4	2
5	One of the key decisions for procurement function is choosing/selecting a supply partner and particularly for those who handle critical elements of supply chain. Discuss the criteria for supplier selection excluding Cost and Quality.	10	4	4

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Department of Aerospace Engineering

Date: 06/06/25	**Test - 2**	**Max. Marks: 60**
Semester: VI	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
Course Title: Fundamentals of Aerospace Engineering		**Course Code: AS266TEA**

S No	Questions	M	BT	CO
1	Among the various gas turbine engine derivatives, _____ engine has the lightest specific weight.	01	L1	1
2	Achieving thrust reversals is quite easy in case of _____ engines.	01	L2	2
3	_____ aids in decreasing the speed and increasing the torque of the propeller shaft connecting the propeller and the turbine of the engine.	01	L2	3
4	Fluid Friction is one of the major factors contributing towards the increase of _____ in an Ideal Brayton cycle.	01	SL2	2
5	_____ is an aerodynamic force acting perpendicular to the chord of a cambered airfoil.	01	L3	2
6	_____ is defined as the location on an airfoil where the distribution of forces results in zero moment.	01	L2	1
7	To get additional lift during take-off, the pilot needs to deploy _____ of an airplane.	01	L2	2
8	Define aerodynamic center of a wing.	01	L2	4
9	What is meant by by-pass ratio of a turbofan engine.	01	L2	4
10	List the various types of drag experienced by a airplane in flight.	01	L2	3
	PART B			
1	Depicting the various components of a pure turbojet engine on neat sketch, explain its working principle. Also, list the advantages and disadvantages of the turbojet engine.	10	L1	CO2
2	Illustrating a neat sketch of a gas turbine engine, explain the operation of Brayton cycle.	10	L3	CO3
3	Define stalling. With a neat sketch, explain the physical phenomenon of the occurrence of stall with respect to the airfoils angle of attack.	10	L2	CO2
4	An aircraft has an effective wing area of $78m^2$. It is in level flight at a speed of 750km/h at an altitude where the density is $0.467kg/m^3$. If the weight of the aircraft is 495kN. Find the coefficient of lift for the wing section. If the coefficient of drag is 10% of the C_L. Determine the thrust required for level flight.	10	L3	CO4
5	Draw a neat diagram of a typical wing planform geometry. Highlight all the parts of the wing and explain.	10	L3	CO3

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Go, change the world®

Academic year 2024-2025 (Even Sem)

Common to all branches

DEPARTMENT OF
BIOTECHNOLOGY

Date	**June, 2025**	**Maximum Marks**	**10 + 50**
Course Code	**BT266TEB**	**Duration**	**120 min**
Sem	**VI Semester**	**Closed Book Offline Quiz and Test – 2**	
Healthcare Analytics (Institutional Elective)			

Answer all questions in Part A & Part B, Answer Part A in first two pages in sequence only

Part A

S No	Questions	M	BT	CO
1	Multiple sequence analysis consists _____ or more number of sequences.	1	2	1
2	Needleman_Wunsch algorithm can be applied for _____ alignment.	1	2	1
3	A phylogenetic tree represents _____	1	2	2
4	Among the PAM1 and PAM250, which one has the more distant?	1	3	2
5	Differentiate between the opening and extension gap penalties.	2	2	3
6	What is null hypothesis?	2	4	3
7	Mention any 2 measures of statistical significance.	2	2	4

Part B

S No	Questions (Test)	M	BT	CO
1.	Illustrate on the various types of sequence alignments. Explain any one in brief.	10	2	3
2	How does the scoring matrix helps to derive the interpretation among sequence alignment?	10	3	3
3	Elucidate on the profiles and Hidden Morkov Models for bioinformatics data.	10	4	4
4	Differentiate between PAM and BLOSSUM and write its significance.	10	4	4
5	Write an overview of molecular phylogeny.	10	2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks			60	40		40	20	40		

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Department of Chemical Engineering

Date: 04.06.2025	**Test - 2**	**Max. Marks: 60**
Semester: 6	**UG**	**Duration: 2 Hrs**
Course Title: Industrial Safety Engineering		**Course Code: CH266TEC**

S No	Questions	M	BT	CO
1	What is the certainty coefficient if cash outflow to a project is 100000 and certain cash flow is 50000?	02	03	02
2	Determine Severity Rate with two recordable injuries, if an employee breaks leg on a Monday, and loses the rest of that day plus three additional days of work, then the employee comes back to work the following Friday and is given restricted work tasks, and then loses another two days.	02	03	02
3	Expand RADR and OIR	02	02	02
4	Write process hazard worksheet table that is used in PHA analysis.	02	02	2
5	Write Fault tree event symbols for basic event and underdeveloped event.	02	02	2
	PART B			
1	What is Fault Tree? Construct the Fault and Event trees for the process consist of reactor system with heating and cooling arrangement to maintain constant reactor temperature.	10	03	03
2	What is Preliminary hazard analysis? Classify and explain preliminary hazard analysis table for FMEA, HAZOP and operator related hazards.	10	02	02
3	Draw a flowchart and explain the basic concept of Safety Domain Ontology.	10	01	03
4	A project is required to invest INR. 1,10,000 and is expected to generate cash flow after tax over its economic life time of 5 years of INR 20,000, INR30,000, INR35,000, INR 55,000 and INR 10,000. Risk free interest rate is 7 per cent and decision maker interested to add 3 per cent as risk premium for the project. Calculate NPV using RADR approach and suggest on the acceptability of the project.	05	03	03
5	The following are the possible net cash flows of Project X and Y and their associated probabilities. Both projects have a discount rate 10 per cent. Calculate the expected NPV for each project if initial investment in both cases is 5000. Calculate variance, standard deviation and coefficient of variation and comment which project is preferable?	15	03	04

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Department of Computer Science and Engineering

Date: June 2025	Test - 2	Max. Marks: 60
Semester: VI	UG	Duration: 2 Hrs
Course Title: ROBOTIC PROCESS AUTOMATION		Course Code: CS266TED

PART A

S No	Questions	M	BT	CO
1.	List the actions that cannot be recorded in UiPath	02	2	1
2.	What is an UI element?	01	2	1
3.	To extract information from virtual environment _____ output method is used	01	1	1
4.	Define data scraping.	01	1	2
5.	What is relative scraping? How does it work?	02	2	1
6.	Outline different types of user interfaces in UiPath.	02	2	2
7.	_____ activity is used to extract data from the DataTable.	01	1	1

PART B

S No	Questions	M	BT	CO
1. a	Discuss different types of recorders in UiPath. Explain any two.	06	2	2
b	Describe copy activity of the recorder in UiPath.	04	3	2
2. a	List the various input methods supported in UiPath and discuss the key differences between them.	06	2	3
b	Identify the steps of data scraping wizard.	04	3	3
3. a	Describe in detail OCR output method provided in UiPath.	04	3	3
b	Explain various activities related to Data Scraping.	06	3	2
4. a	Elaborate on various types of selectors available in UiPath.	04	2	2
b	Analyse the usage of the UI Explorer with an appropriate example.	06	3	2
5. a	Name and explain the processes involved in Retrieving Information.	04	4	3
b	What are anchors? Describe its usage in PDF documents with a workflow.	06	4	4

Course Outcomes

CO1	Understand RPA principles, its features and applications
CO2	Demonstrate proficiency in handling variables and decision making inside a workflow and data manipulation techniques.
CO3	Gain insights into recording, email automation and exception handling and orchestrator
CO4	Analyze the trends in automation and choose business strategy to design a real world automation workflow.

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

BT & CO Marks	L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4	CO5
	3	23	24	10	–	–	7	33	14	6	–

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Department of Civil Engineering

Date:	CIE - 2	Max. Marks: 60
Semester: VI	UG	Duration: 2 Hrs
Course Title: Intelligent Transportation System		Course Code: CV266TEE

Quiz

S No	Questions	M	BT	CO
1.	List any two functionalities required for ITS user services.	2	2	1
2.	Mention two methods of data acquisition in ITS.	2	2	1
3.	Explain the importance of ITS physical architecture.	2	2	2
4.	Mention any two technology building blocks of ITS.	2	2	1
5.	Describe the role of communication tools in ITS.	2	2	2

Test

S No	Questions	M	BT	CO
1.	Analyze the differences between logical and physical ITS architectures.	10	2	1
2.	Analyze the roles of detection, identification, and collection methods in ITS performance.	10	3	1
3.	Examine the contribution of each ITS building block in improving urban transportation efficiency.	10	2	1
4.	Discuss the importance and limitations of ITS standards in national implementations.	10	3	2
5.	Propose a detailed ITS evaluation framework for a newly implemented smart city transport project.	10	5	2

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Department of Chemistry

Date: 04-06-2025	**Test - 2**	**Max. Marks: 50**
Semester: 6	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
Course Title: ADVANCED ENERGY STORAGE FOR E-MOBILITY		**Course Code: CM266TEG**

S No	Questions	M	BT	CO
1	Apply your understanding of sodium-based battery chemistries to explain different types of sodium batteries. Using a neat and labelled diagram, describe the construction and working of a sodium-sulfur (Na-S) battery along with advantages in safety point of view.	10	3	4
2	Outline the construction and working of advanced lead acid battery along with chemical reactions involved during charging and discharging. Mention its advantages.	10	2	1
3	Compare the electrochemical performance of lithium iron phosphate (LFP) with that of lithium cobalt oxide battery. Explain the working mechanism of Lithium iron phosphate battery along with neat labelled diagram.	10	4	3
4	Illustrate the construction and working of lithium-sulfur batteries and discuss their advantages and limitations. Comment on the polysulfide shuttle mechanism in such type of battery.	10	3	2
5	Explain how the ZEBRA (Sodium Nickel Chloride) battery operates and its suitability for commercial applications. Discuss the advantages and limitations of zebra battery.	10	2	1

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Department of Chemistry

Date: 04-06-2025	**Quiz - 2**	**Max. Marks: 10**
Semester: 6	**UG**	**Duration: $\frac{1}{2}$ Hrs**
Course Title: ADVANCED ENERGY STORAGE FOR E-MOBILITY		**Course Code: CM266TEG**

S No	Questions	M	BT	CO
1	Give an example for the cathode materials used in lithium sulphur battery.	1	1	
2	Identify one difference between sodium-ion and lithium-ion batteries.	1	2	
3	Name the electrolyte used in lithium-sulphur batteries.	1	2	
4	Illustrate with one example how a battery's chemistry affects charging speed.	1	3	
5	Name the material which deposits during charging and discharging of lead acid battery.	1	1	
6	Lithium battery has very high capacity than that of lead acid battery. justify	1	5	
7	Justify why sodium-sulphur batteries are not commonly used in consumer electronics.	1	5	
8	Suggest an appropriate non-lithium battery for grid storage applications.	1	3	2
9	Analyse the main limitation of lithium-air batteries in practical applications.	1	5	4
10	Apply your knowledge to name one battery suitable for low-temperature EV operations.	1	3	2

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Department of Electronics and Communication Engineering

Date: 4.6.2025	**Test - 2**	**Max. Marks: 50**
Semester: VI	**UG**	**Duration: $1\frac{1}{2}$ Hrs $+\frac{1}{2}$ Hrs**
Course Title: Human Machine Interface		**Course Code: EC266TEH**

S No	Questions	M	BT	CO
1.a.	List any FOUR differences between UX and UI.	4	1	1
1.b.	Discuss the core components of UX designing applications.	6	2	1
2.a.	Explain the advantageous of Mobile HMI for industrial applications.	4	2	1
2.b.	List and explain the 5 elements of UX in detail with neat diagrams.	6	2	2
3.a.	Discuss the key roles of web server in HMI Application.	5	2	2
3.b.	List the main stages of the user-cantered HMI development process and explain the importance of each stage.	5	2	2
4.a.	Illustrate the Design Thinking process in detail for HMI applications.	6	2	3
4.b.	Explain five common User Experience asset design types.	4	2	1
5.a.	Discuss the 5 Dimensions of Interactive design used for SCADA Applications.	5	2	2
5.b.	What is a web server? Describe in brief the main components of a web server.	5	1	1

S No	Quiz -II	M	BT	CO
1.	Cross-platform application programming interface (API) for rendering 2D and 3D vector graphics is called _____	1	1	1
2.	The style and formatting of text used in a product, including font families, sizes, and weights is called _____	1	1	1
3.	Simple sketches or digital mockups that show the layout, structure, and organization of content on a webpage or app screen are called _____	1	1	1
4.	Fictional characters that represent the target users of a product, used to guide design decisions and keep the user's needs in mind are known as _____	1	1	1
5.	Name any 2 prototyping tools for UX.	1	1	1
6.	Name two accessibility features that should be considered in a mobile HMI.	1	2	1
7.	_____ protocol is most commonly used for communication between a webserver and a browser-based HMI.	1	2	1
8.	Which of the following is considered a fundamental UI design component?	1	1	2
9.	_____ file format is most commonly used for vector design assets.	1	1	1
10.	The main benefit of using vector assets in design _____	1	1	1

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 4th June 2025	**CIE– II**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Elements of Financial Management (Institutional Elective)		**Course Code : IM266TEQ**

Note:
1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List the three commonly used approaches employed for valuation of equities.	02	L3	CO2
2.	Distinguish between Market Risk and Unique Risk.	02	L2	CO2
3.	In the context of long- term financing, distinguish between Equity and Debt in terms of the ownership rights to the investor.	02	L2	CO4
4.	What does the term Yield to Maturity represent?	01	L2	CO2
5.	What does the term "Risk" in Finance primarily refer to?	01	L1	CO3
6.	Mention any one capital budgeting technique that considers the time value of money.	01	L1	CO3
7.	Name the capital budgeting technique that calculates profitability as a percentage of investments.	01	L1	CO3
	Part – B			
1 a.	A bond with a face value of ₹1,000 carries a coupon rate of 10% and matures in 5 years. If the required rate of return is 8%, calculate the value of the bond.	05	L3	2
b.	A bond with a face value of ₹1,000 sells for ₹950. It has a 10% annual coupon and matures in 5 years. Calculate the Yield to Maturity.	05	L3	2
2 a.	An investor is analysing a stock with the following possible returns and probabilities: Scenario / Return (%) / Probability Boom / 20 / 0.3 Normal / 10 / 0.5 Recession / -5 / 0.2 Calculate the Expected Return and also the Standard Deviation of the returns.	07	L3	2
b.	Relationship between risk and Return is one of the fundamental Principles in Financial decisions in organizations. Justify	03	L2	2
3.	Enumerate the steps involved in Capital Budgeting Process.	10	L2	3

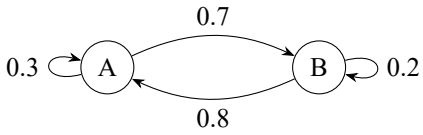
4 a.	There are two Projects for consideration. Apply Net Present Value method and determine the Project to be chosen. Project A: ₹20,000 investment, ₹6,000 for 5 years Project B: ₹20,000 investment, ₹10,000 for 2 years Rate = 10%	05	L4	3
b.	A project costs ₹150,000 with no residual value. It yields accounting profits of ₹10,000, ₹12,000, ₹14,000, ₹16,000, and ₹18,000 over 5 years. Calculate Average Rate of Return and interpret.	05	L4	3
5.	Explain any Five sources of long term finance available to business organizations.	10	L2	4

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Department of Mathematics

Date: 04/06/2025	**CIE - 2**	**Max. Marks: 100**
Semester: VI	**UG**	**Duration: 2 Hrs. (2pm-4pm)**
Course Title: Mathematical Modelling		**Course Code: MA266TEU**

SL. No	Quiz: Questions	M	BTL	CO
1	The Fibonacci Numbers 1, 1, 2, 3, 5, 8, 13, 21,…can be modelled in a difference equation of the form _____.	2	2	2
2	Solve $u_{t+1} = 8x_t$, $x_0 = \frac{1}{2}$.	2	1	1
3	With respect to a characteristic brown eyes, suppose the proportions of the dominants, hybrids and recessives in 9th generation are p_9, q_9 and r_9 respectively, then r_{10} = _____, and q_{10} = _____.	2	1	1
4	In the long run, the proportion of time that the given Markov chain will be in state A is _____ and state B is _____. 	2	3	2
5	Find the fixed probability vector of the stochastic matrix $A = \begin{bmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{bmatrix}$.	2	3	3
SL. No	**Test**	**M**	**BTL**	**CO**
1.	Develop a model through difference equations in genetics based on Hardy-Weinberg law.	10	2	2
2.	Discuss Cobweb model about the variation of price and quantity of a commodity produced by a farm.	10	2	2
3.	Prove that the Markov chain with transition probability matrix $\begin{bmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix}$ is irreducible. Also find its stationary probability vector.	10	3	2
4. a	Suppose a person borrows a sum S_0 at compound interest i and wants to 'amortize' his debt and S_t be the amount due at the end of t years. Model this in the form of a difference equation and find the amortization factor.	6	3	3
4. b	A college professor drives or takes the college bus to reach the college. If he drives to work, then the next day he takes the bus with the probability 0.2. On the other hand if he takes the bus, then the next day he drives with probability 0.3. Find the transition matrix for the chain of mode of transport.	4	3	3

5	Consider a network containing five individuals viz. A, B, C, D and E. The system A can communicate to B and C; B can communicate to A and C; C can communicate to A, B and D; D can communicate to C and E; and E can communicate to B only. Assuming that if a person can communicate with n people, he will send a message to any one of them with probability $1/n$, rank the five people in accordance with their importance in the network in the long run.	10	4	3

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Department of Mathematics

Date: 04-06-2025	**CIE 2**	**Max. Marks: 10+50**
Semester: VI	**UG**	**Duration: 2 Hrs (2PM-4PM)**
Course Title: Mathematics of Quantum Computing		**Course Code: MA266TEV**

S No	Quiz	M	BT	CO
1.	How many queries does the Deutsch-Jozsa algorithm require to determine whether a function is constant or balanced? What is the co-domain of the function involved?	2	1	1
2.	In the Bernstein-Vazirani problem, we are given a function $f(x) = a \cdot x \; mod 2$. What does the algorithm help us find? How many Hadamard gates are utilized in it?	2	1	1
3.	The Phase gate S modifies which component of a qubit? What is the modification?	2	2	2
4.	In the Deutsch-Jozsa algorithm, why is quantum parallelism crucial?	2	3	3
5.	What mathematical technique is used in Simon's algorithm to extract the hidden string s?	1	2	3
6.	Which gate sequence can create the Bell state $\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$?	1	2	2
Sl No	**Test**	**M**	**BT**	**CO**
1.	Use the Z-Y decomposition technique to factorize the unitary $U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$ into a sequence of rotations around the Z and Y axes, including any overall phase factor.	10	2	1
2.	Describe the Deutsch Algorithm and its purpose. For a single-qubit oracle function $f : \{0,1\} \to \{0,1\}$, derive the final state after applying the Deutsch circuit, starting from $\|0\rangle\|1\rangle$, and explain how the measurement distinguishes between constant and balanced functions.	10	3	3

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DEPARTMENT OF
HUMANITIES AND SOCIAL SCIENCES

Date : 04.06.2025	**CIE– II**	**Max. Marks : 10 + 50**
Semester : VI	**UG- Institutional Elective**	**Duration : 30 + 90 Min**
Course Title: Applied Psychology for Engineers		**Course Code : HS266TEW**

Note:
Answer all the Questions

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List the characteristics of Intelligence Tests.	2	L1	CO1
2.	What is the main purpose of an aptitude test?	2	L2	CO2
3.	Define the concept of reciprocal determinism in Bandura's social-cognitive theory of personality.	2	L2	CO3
4.	How do people high in self-esteem differ from those low in self-esteem?	2	L2	CO4
5.	What is Frustration? Mention the important sources of Frustration.	2	L1	CO3
	Part – B			
1	A 12-year-old boy is performing poorly in school despite appearing attentive and hardworking. His teachers believe he might have a learning disability, but his parents think he is just not interested in academics. The school counselor decides to administer an intelligence test. What type of intelligence test would be most appropriate for the boy's case—verbal or non-verbal? Why?	10	L2	CO1
2	A software company wants to hire interns for a programming role. To ensure the best candidates are chosen, they design an assessment that includes a programming aptitude test. Explain how aptitude tests can be standardized to ensure fairness in employee selection.	10	L2	CO2
3	Seth constantly struggles between doing what he wants and what he believes is "right." He often feels guilty even after small personal indulgences and describes a persistent inner conflict that drains his energy. Explain what happens if there is an imbalance among the three structures of personality.	10	L3	CO2
4	A 21-year-old college student named Beck is facing difficulty in choosing a career path. He says he feels "lost" and cannot make decisions confidently. A clinical psychologist chooses to administer the Thematic Apperception Test (TAT). How can the TAT help uncover underlying aspects of Beck's personality that may be influencing his indecisiveness?	10	L2	CO2
5	James, a 35-year-old marketing executive, has been working overtime for months due to a major project deadline. He notices increased anxiety, irritability, and stomach problems. His family complains he's emotionally distant. Explain how potential sources of psychological stress might be affecting James' health.	10	L4	CO2

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Department of Civil Engineering

Date	**6th June 2025**	**Maximum Marks**	**50+10**
Course Code	**HS266TEY**	**Duration**	**90+20 Min**
Sem	**VI**	**CIE – I (Test)**	
Universal Human Values – III			

Sl. No.	Quiz	M	BT	CO
1	The need for continuous happiness is fulfilled by _____ & _____.	02	1	1
2	Human being is the coexistence of _____ and _____.	02	1	1
3	Enlist any two higher activities of the self.	02	1	2
4	Name any two examples of physical order and bio order	02	2	2
5	Temporary happiness is to _____ Continuous happiness is to _____.	01	2	1

Sl. No.	Test	M	BT	CO
1	Write a detailed note on innateness, natural characteristics and inheritance of four orders in nature.	10	1	1
2	(i) Explain the sources of continuous happiness and temporary happiness with an example. (ii) Which are the three bases of selecting, tasting is satisfied in mother's cooked food?	5	2	2
		5	3	2
3	Explore the role of the human being as the knower, experiencer and doer in the domain of right understanding.	10	3	3
4	Elaborate the impact of right understanding on the conduct of human being.	10	3	3
5	Discuss how the higher activities of the self-guides the lower activities and brings the harmony at all levels.	10	2	2

Course outcomes:

CO 1	Understand the basic human aspiration with program of its fulfillment and meaning of resolution in the complete expanse of human living
CO 2	Understand human being in depth and see how self is central to human being
CO 3	Understand existence in depth and see how coexistence is central to existence
CO 4	Understand human conduct and the holistic way of living leading to human tradition

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max	16	24	20	–	16	19	25	–	–	–

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Academic year 2023-2024 (EVEN Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**24th July 2024**	**Maximum Marks**	**10**
Course Code	**21IE6F8**	**Duration**	**30 Min**
Sem	**VI**	**CIE – II**	
Elements of Financial Management (Institutional Elective)			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	In the context of bond valuation, what does the term Yield to maturity refer to?	1	2	CO1
2	_____ type of bond pays no periodic interest but is issued at a discount.	1	1	CO2
3	The model used to determine the expected return on a stock based on its risk relative to the market is _____.	1	1	CO2
4	Name the valuation method that calculates the present value of a project's cash flows minus its initial investment.	1	1	CO1
5	What is the term for the risk associated with an entire market or market segment?	1	2	CO1
6	_____ refers to a strategy that involves spreading of investments to reduce risk.	1	1	CO1
7	What is the term that describes the potential for loss in an investment?	1	2	CO1
8	Mention the capital budgeting technique that measures the profitability of a project by calculating the return rate that makes the NPV equal to zero.	1	1	CO2
9	The period required for the return on an investment to repay the original investment is _____.	1	1	CO2
10	_____ method evaluates the efficiency of an investment by comparing the accounting profit to the initial investment cost.	1	1	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	05	05	–	–	07	03	–	–	–	–

Academic year 2023-2024 (EVEN Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**24th July 2024**	**Maximum Marks**	**50**
Course Code	**21IE6F8**	**Duration**	**90 Min**
Sem	**VI**	**CIE – II**	
Elements of Financial Management (Institutional Elective)			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – B			
1a	A Rs.200/-par value bond carries a coupon rate of 12% and maturity period of 8 years. The interest rate is payable semi-annually. Compute the value of the bond if Required Rate of Return is 14%.	5	3	CO1
b	The expected dividend per share is Rs.12/-. The dividend is expected to grow at the rate of 7% per year. If the price of the share is Rs.145/- now, determine the Expected rate of Return.	5	3	CO1
2a	The current dividend for an equity share is Rs.6/-. If the constant compound growth rate is 6%, determine the dividend 10 years hence.	5	3	CO1
b	The price per bond of M/S. Akshara Industries is Rs.180/-. The par value of the share is Rs.200/-. The coupon rate of the bond is 14%. The maturity period for the bond is 6 years. Determine the Yield to Maturity for the investor.	5	3	CO1
3	The stock of Box Limited performs well relative to other stocks during recessionary periods. The stock of Cox Limited, on the other hand, does well during growth periods. Both the stocks are currently selling for ₹ 100 per share. You assess the rupee return (dividend plus price) of these stocks for the next year as follows:	10	4	CO1

	Economic Condition			
	High Growth	**Low Growth**	**Stagnation**	**Recession**
Probability	0.3	0.4	0.2	0.1
Return on Box's Stock	100	110	120	140
Return on Cox's Stock	150	130	90	60

Calculate the expected return and standard deviation of investing:
(i) ₹ 1,000 in the equity stock of Box Limited
(ii) ₹ 1,000 in the equity stock of Cox Limited

4a	Enumerate the steps involved the process of Capital Budgeting.	5	2	CO2

Academic year 2023-2024 (EVEN Sem)

b	A project which is being evaluated by a firm has a cost of capital of 12%. The initial outlay for the project is Rs.2 lakhs. The benefits accrued for the next four years are given below. Year | 1 | 2 | 3 | 4 Amount (Rs) | 50,000 | 80,000 | 80,000 | 1,00,000 Calculate the Benefit Cost Ratio and Net Benefit Cost Ratio for the project. Also based on the standard decision criteria, suggest the feasibility of investing in the project.	5	4	CO2
5	Long-term finance is critical for a company's growth and stability, providing the necessary capital to fund substantial investments and strategic initiatives. In the light of the above statement, provide a brief review of different types of sources.	10	2	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	30	10	10	–	–	15	20	15	–	–